MILITARY SPECIFICATION

CLOTH, COATED AND LAMINATED, POLYCHLOROPRENE ON NYLON, AND TAPE, POLYCHLOROPRENE, UNSUPPORTED

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for the procurement of nylon base polychloroprene coated cloth and laminated cloth and polychloroprene unsupported tape intended for use in the insulation garment and continuous wear anti-exposure coveralls designated as Mark 5.

1.2 Classification - The cloth and tape shall be of the following types and classes as specified (see 6.2):

Type I - Cloth, Coated, Rubber, Knitted Stretch Nylon

Type II - Cloth, Laminated, Rubber, Neck and Wrist Seal

Type III - Tape, Rubber, Unsupported

Class 1 - Pressure Sensitive Tape

Class 2 - Non-pressure Sensitive Tape

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

FSC 8305

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SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods
PPP-P-51	Packaging, Packing and Marking of Textile Fabrics (Woolen, Worsteds, Cottons, Silks and Synthetics)

STANDARDS

Federal

FED-STD-601	Rubber: Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-654	Webbing, Packaging and Packing of

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by the contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code DCI), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Color Association of the United States, Inc.

TCA Cable Numbers and Colors

(TCA publications may be obtained from the Color Association of the United States, Inc., 200 Madison Avenue, New York 16, New York.)

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3. REQUIREMENTS

3.1 Materials - The finished coated cloth and laminated cloth shall consist of knitted nylon base cloths uniformly coated with a polychloroprene compound. The unsupported tape shall consist of the cured polychloroprene compound protected with a suitable liner.

3.1.1 Base cloth - The base cloth shall be a dyed, plain jersey circular knit made of nylon stretch yarns. The yarns shall be nylon polyamide of polyhexamethylene adipamide. The color of the base cloth shall match TCA Cable No. 70072 (crepe side), Indian Orange (see 6.3).

3.1.2 Coating - The coating compound tested as specified in 4.4.5 shall contain not less than 60 percent by volume of polychloroprene. The remainder of the compound shall consist of softeners, curing agents, anti-oxidants, and reinforcing materials. The coating shall be the natural color of polychloroprene or suitably pigmented during compounding so that the cured compound and base cloth of the coated and laminated cloths shall have a matching (Indian Orange) color. The coating compound shall be compatible with the base cloth and contain no waxes or other ingredients that may bloom to the surface to adversely affect the coating adhesion, and cementability of the finished cloth. The ingredients of the compound shall be water insoluble after curing.

3.2 Application of coating - After application of a foundation coat, which must be compatible with the base cloth and the remainder of the coating compound to produce the adhesion specified, the coating compound shall be applied by a spreader or calender coating operation. The coating shall be applied to one surface, the side with a smooth surface to provide the best coating adhesion on Type I coated cloth and applied between two cloths on Type II laminated cloth.

3.3 Curing - The coated cloth and laminated cloth shall be cured to meet the performance requirements of the specification. The cured polychloroprene coating may be lightly dusted with talc or zinc stearate.

3.4 Properties of coated cloth and laminated cloth - The coated cloth and laminated cloth properties shall conform to the requirements specified in Table I and in the following paragraphs when tested as specified therein.

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TABLE I

Properties of Coated Cloth and Laminated Cloth

CHARACTERISTIC	SPECIFICATION CCC-T-191 METHOD NO.	REQUIREMENTS	
		TYPE I COATED CLOTH	TYPE II LAMINATED CLOTH
Weight, ozs/sq yd, max.	5041	19.0	24.0
Breaking strength, 2 inch wide strip, lbs, min.	5102.1		
Wales		90	90
Courses		45	45
Elongation at break, 2 inch wide strip, percent, min.	5102.1		
Wales		100	100
Courses		250	250

3.4.1 Coating thickness - In each type of cloth the coating shall be not less than 0.012 inch thick (see 4.4.6).

3.4.2 Adhesion - The coating adhesion to the base cloth Type I coated cloth and the ply adhesion of Type II laminated cloth shall be not less than 8 pounds for a 2-inch specimen when tested in the course direction (see 4.4.7).

3.4.3 Modulus at 75 percent elongation - For each type of cloth, the load required to maintain 75 percent elongation on a 2-inch wide specimen, in the course direction, after the specimen has previously been stretched 150 percent, shall not exceed 3.0 pounds (see 4.4.8).

3.4.4 Resistance to flexing - The Type I cloth shall be subjected to 1000 flexing cycles at 50 percent stretch in the course direction and Type II cloth shall be subjected to 500 flexing cycles at 100 percent stretch in the course direction. After flexing, the specimen of each type of cloth shall have no tears in the coating, no separation of coating from the cloth, nor shall it leak water (see 4.4.9).

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3.4.5 Resistance to low temperature exposure - After low temperature exposure for 30 minutes at -65 ± 2°F (-54 ± 1.1°C), the specimen of each type of cloth shall show no signs of cracking, and it shall not leak water (see 4.4.10).

3.4.6 Tension set - After being stretched to 100 percent for 16 hours, the tension set shall not exceed 13 percent when examined 3 hours later (see 4.4.11).

3.5 Dimensions of the coated cloth and laminated cloth -

3.5.1 Length - The coated cloth and laminated cloth shall be furnished in rolls. The minimum length of a roll, the maximum number of pieces in a roll, and the minimum length of the shortest piece in a roll shall be as specified by the procuring activity (see 6.2).

3.5.2 Width - The coated cloth and laminated cloth shall be furnished in the width specified by the procuring activity (see 6.2).

3.6 Unsupported tape - The Type III unsupported tape shall be made from polychloroprene conforming to the coating compound requirements specified in 3.1.2. The color of the unsupported tape shall be an approximate match with the cloth (Indian Orange) used in Types I and II.

3.6.1 Class 1 pressure sensitive tape - The pressure sensitive tape shall have a surface coating of a polychloroprene content uncured or partially cured, 2.3 to 3.8 ounces per square yard in weight, and covered with a protective lining. The cured side shall be lightly dusted with talc.

3.6.2 Class 2 non-pressure sensitive tape - The non-pressure sensitive tape shall be cured in the same manner and degree of cure as the cloth of this specification. One side of the tape shall have a protective liner and the other side shall be lightly dusted with talc.

3.6.3 Width - The width of the tape shall be as specified by the procuring activity (see 6.2) and the thickness shall be 0.015 +0, -0.003 inch.

3.6.4 Unless otherwise specified by the procuring activity, the tape shall be furnished in rolls of 72 ± 2 yards continuous length with splices. Additional roll put-up requirements may be specified by the procuring activity.

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3.7 Workmanship - The coated cloth, the laminated cloth, and the unsupported tape shall be uniformly processed to produce the high quality products established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Compliance - Determination of compliance with provisions of this specification shall include quality conformance (lot-by-lot) inspection.

4.2.1 Quality conformance (lot-by-lot) inspection - The quality conformance inspection shall consist of examining and testing samples of coated cloth and laminated cloth from each lot for conformance to Section 3 of this specification, and an examination of the sample of filled shipping containers from each lot for conformance to the packaging, packing, and marking requirements.

4.3 Sampling and acceptability criteria - Quality conformance inspection samples shall be taken from each lot of coated and laminated cloth and tape. A lot shall consist of coated and laminated cloth and tape made under essentially the same conditions and in a single production run. Unless otherwise specified, the lot size shall be expressed in units of 1 yard each.

4.3.1 Samples for yard-by-yard examination - The unit of product for this examination shall be one linear yard. A sample size for the yard-by-yard examination shall be selected in accordance with Standard MIL-STD-105, Inspection Level II, with an acceptable quality level of 2.5 for major defects and 6.5 for minor defects per 100 units (yards). The required sample yardage to be examined shall comprise an approximately equal number of yards out of each selected roll.

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4.3.2 Sample for overall examination and dimension examination - The unit of product for these examinations shall be one roll. Examination shall be in accordance with Standard MIL-STD-105, Inspection Level S-2, Acceptance Number zero. The minimum sample size shall be three rolls.

4.3.3 Sample for tests - The unit of product shall be 2 yards of the coated cloth and laminated cloth. Testing (Table I) shall be in accordance with Standard MIL-STD-105, Inspection Level S-2, Acceptance Number zero. The minimum sample size shall be two units. The test sample shall include four 6 by 6 by .07 to .08 inch sheets of cured coating compound from each batch and the coating compound samples shall be identified with the lot number of the cloth to which that batch is applied as well as with the batch number. The test samples of coating compound and laminated cloth, prior to application of coating, shall have been cured to the same degree of cure. The lot shall be unacceptable if one or more units fail to meet any requirement specified.

4.3.4 Sample for packaging, packing, and marking examination - The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The sample size shall be determined in accordance with Standard MIL-STD-105, Inspection Level S-2, and the Acceptable Quality Level shall be 4.0 percent defective.

4.4 Inspection methods - Unless otherwise specified, inspection tests (of Table I) shall be conducted in accordance with the provisions of Standard MIL-STD-105. Test conditions and methods shall be as described in Specification CCC-T-191, wherever applicable. Unless otherwise specified, the physical and chemical requirement values specified in Section 3 apply to the average of the determinations made on a unit of the product. Samples shall be tested in accordance with the methods listed in Table I.

4.4.1 In-process material examination - The examination will be made at any point or any phase of the manufacturing process to determine whether the material and workmanship is as specified in Section 3.

4.4.2 Yard-by-yard examination - In each roll of cloth selected, an approximately equal number of yards comprising the sample shall be subjected to the examination. The defects found shall be identified and classified in accordance with Tables II and III, as applicable. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs.

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TABLE II

Classification of Defects for Coating

DEFECTS	CLASSIFICATION	
	MAJOR	MINOR
Blisters or lumps		
- 1/4 square inch or less in combined directions, clearly visible 1/		X
- Over 1/4 square inch in combined directions, clearly visible 1/	X	
Coating faults		
- Blooming	X	
- Creases	X	
- Folds - winding on tubes		X
- Uneven coating		
- Heavily coated areas, clearly visible 1/		X
- Thin areas where coating compound is insufficient	X	
Coating missing	X	
Cuts	X	
Pinholes	X	
Foreign matter incorporated in coating	X	
Splice mark clearly visible on coated side 1/		
- If coating is not removed		X
- If coating is missing	X	
Stains in coating surface		
- Any spot or stain (compound, oil, dirt) over 1 inch in length in any direction, clearly visible 1/		X
Steaks		
- Shade streaks in coating, clearly visible 1/		X
Tears	X	

1/ The term "clearly visible" shall be interpreted as meaning clearly visible at normal examination distance (approximately 3 feet).

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TABLE III

Classification of Defects for Base Cloth

DEFECTS	CLASSIFICATION	
	MAJOR	MINOR
Up to 2 knots per square yard of (relaxed) cloth		X
Over 2 knots per square yard of cloth	X	
Holes	X	
Runs	X	
Shade Bars	X	
Drop Stitches	X	
Snags or Pulls	X	
Spots (oil or dirt)	X	
Stains	X	

4.4.3 Overall examination - The required number of pieces of coated and laminated cloth and tape in the sample (see 4.3.2) shall be examined for defects as listed in Table IV. Each defect listed in Table IV shall be counted no more than once in each roll examined.

TABLE IV

Overall Defects

Selvage curled, folded, rolled, or slack continuously or intermittently throughout the piece.
Overall uncleanness or objectionable odor.
Uneven coating throughout the piece.
Off-shade, shaded end to end, side to side, side to center, or throughout the piece.
Mottled, cloudy, or streaky throughout the piece.
Date of coating application missing from end of roll or identification tag.

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4.4.4 Dimensions examination - Each piece of coated and laminated cloth and tape in each roll in the sample (see 4.3.2) shall be counted and measured. Width and length shall be measured as specified in Methods 5020 and 5010 of Specification CCC-T-191. Any of the following dimensions of the sample found to be less than the minimum dimensions specified for a roll in the contract shall be considered a defect; namely:

(a) Gross length of roll

(b) Width of roll

(c) Length of shortest piece on roll

When the number of pieces per roll exceed that specified in the contract, this shall also be considered a defect. The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on piece tickets. The lot shall be unacceptable if the width of the coated and laminated cloth and tape is not within tolerances specified in the contract (see 6.2).

4.4.5 Polychloroprene compound - The volume of polychloroprene present in the coating compound shall be determined in accordance with Method 15111 of Standard FED-STD-601.

4.4.6 Coating thickness - The coating thickness shall be determined in accordance with Method 2011 of Standard FED-STD-601. The coating film shall be physically removed from the cloth. When the coating film cannot be removed from the cloth, the thickness of the coating shall be determined using coated cloth thickness minus cloth thickness.

4.4.7 Coating adhesion - The coating adhesion test specimen shall be subjected to the adhesion test in accordance with Method 8011 of Standard FED-STD-601. Specimens for test may be prepared by use of suitable adhesives.

4.4.8 Modulus - Test specimens, 2 ± 0 by 3 ± 1/2 inches, shall have a 1 inch gage length inscribed and shall be tested in a low capacity tester (20 lb maximum), operated at a speed of 5 inches per minute. The tester shall be operated to stretch the 1 inch gage marks to 2-1/2 inches and then immediately reversed to decrease the gage length to 1-3/4 inches. The number of pounds indicated on the test machine at the 75 percent elongation shall be recorded as the modulus.

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4.4.9 Resistance to flexing - A test specimen of each type of cloth, 8 inches wide and at least 8 inches long in the course direction, shall be marked with a 6-inch gage length in the center of the course direction. The length of the course direction shall be governed by the clamps of the flexing machine. After the specimen has been subjected to the specified flexing cycles at a speed of 10 cycles per minute, the specimen shall be examined for evidence of tears in the coating and separation of the coating from the cloth. The specimen shall then be tested for water leaks in accordance with Method 5516 of Specification CCC-T-191. The water shall be applied for 3 hours at a height of 60 inches for Type I coated cloth and for 3 hours at a height of 36 inches for Type II laminated cloth.

4.4.10 Resistance to low temperature exposure - Resistance to low temperature exposure of a specimen of each type of cloth shall be determined in accordance with Method 5874 of Specification CCC-T-191. The specimen shall be subjected to a temperature of -65 ± 2°F (-54 ± 1.1°C) for a total of 30 minutes. The specimen shall be examined for evidence of cracking. The specimen shall then be tested for water leaks in accordance with Method 5516 of Specification CCC-T-191. The water shall be applied for 5 minutes at a height of 60 inches for Type I coated cloth and for 5 minutes at a height of 36 inches for Type II laminated cloth.

4.4.11 Tension set - A test specimen, identical with the modulus specimens of 4.4.8, shall be maintained at 100 percent elongation in the course direction for 16 hours. The elongation shall be gradually released in 1/2 a minute. After allowing the specimen to recover for 3 hours, the gage length shall be measured and the percent tension set determined.

4.4.12 Packaging, packing, and marking examination - Each sample shipping container fully prepared for delivery shall be examined to determine compliance with the packaging, packing, and marking requirements of this specification. Examination shall be made for the following defects:

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Markings, exterior and interior	Missing; incorrect; illegible; of improper size, location, or method of application.
Material	Component materials nonconforming, missing, damaged, or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, etc. Bulging or distortion of containers.
Weight	Gross weight exceeds requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking - Cloth shall be packaged, packed, and marked in accordance with requirements for Class 2 fabrics of Specification PPP-P-51, except that, prior to rolling, the face of the coated cloth shall be dusted with a suitable talc to prevent tackiness and sticking. Tape shall be packaged, packed, and marked in accordance with Standard MIL-STD-654. The levels of packaging and packing shall be as specified by the procuring activity (see 6.2). In addition to the markings required, each tag shall be marked with the following information: "Date of Application of Coating." In addition, the following precautionary marking shall appear on all containers: "STORE IN A COOL DRY PLACE" (see 4.4.12).

6. NOTES

6.1 Intended use - The cloths and tapes are intended for use in the fabrication of insulation garment and continuous-wear anti-exposure coveralls designated as Mark 5.

6.2 Ordering data - Procurement documents should specify the following:

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SPECIFICATION ANALYSIS SHEET
NAVSEPS FORM 6121/8 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R289

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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